

Does Financial Language Understanding Transfer to Market Prediction? A Systematic Study Across Models, Retrieval Paradigms, and Prediction Targets

Surya Suresh

Dheeraj Gosula

Nikhil Kumar

Abstract

Earnings call transcripts and SEC financial filings are primary sources of financial information, yet it remains unclear whether models that better understand this language produce more reliable predictive signals. We investigate this question through three coordinated research components. The first systematically compares fine-tuning strategies, single-task, multi-task, and curriculum fine-tuning, across three model architecture families on three financial language understanding tasks. The second studies retrieval-augmented generation as an alternative to parametric fine-tuning, comparing sparse, dense, and hybrid retrieval strategies and evaluating whether retrieval can substitute for or complement domain adaptation. The third evaluates predictive transfer: whether language understanding improvements propagate to stock direction, earnings surprise, and volatility predictions, and whether this holds across market periods and industry sectors. Together, these components provide a thorough analysis of financial NLP methods, highlighting when strong language understanding actually translates into meaningful predictive performance.

1 Problem Definition and Research Questions

Earnings call transcripts and SEC annual reports contain financial information expressed in natural language, processed by market participants to form price expectations. Prior NLP work treats financial language understanding and market prediction as separate, leaving unclear whether a model that better extracts and reasons over financial text actually produces better downstream predictions. Our central research question is: *does improved financial language understanding reliably transfer to improved market signal quality, and if so, under what conditions?*

This question is broken down into three sub-questions, each the focus of one research compo-

nent. First, which fine-tuning strategies and model architectures produce the strongest financial language understanding, and does multi-task training help or hurt? Second, is retrieval-augmented generation a viable alternative to fine-tuning for financial QA, and how do retrieval quality and strategy affect end-to-end performance? Third, do language understanding improvements transfer uniformly across prediction targets, market conditions, and document types, or are the gains narrow and task-specific?

2 Tasks, Data, and Evaluation

2.1 Language Understanding Tasks

We evaluate three tasks that capture different aspects of financial language comprehension.

Task 1: Numerical extraction. A model identifies structured financial values (revenue, EPS, guidance figures) and their associated attributes from transcript text. Evaluation uses precision, recall, and F1 against manually verified annotations. We distinguish between exact match and partial credit to capture near-miss errors.

Task 2: Evidence-grounded QA. A model answers factual financial questions and cites the specific transcript spans that support its answer. Evaluation uses answer accuracy, citation exact match, citation partial match, and hallucinated number rate. The hallucination metric is computed by checking whether numerical values in generated answers appear in the source document.

Task 3: Financial sentiment classification. A model assigns positive, negative, or neutral sentiment to financial statements. This serves as a standard benchmark for financial language understanding, enabling comparison with established baselines. Evaluation uses macro-averaged F1.

2.2 Downstream Prediction Tasks

We evaluate predictive transfer on three targets derived from market data. **Stock direction** predicts whether a company’s stock closes up or down in the trading window immediately following an earnings call. **Earnings surprise direction** predicts whether reported earnings per share beat, meet, or miss analyst consensus estimates; this tests whether transcript signals predict firm-specific news rather than just market sentiment. **Post-call volatility** predicts whether realized return volatility in the five trading days following a call falls in the top quartile of the distribution; this tests whether model uncertainty signals capture investor uncertainty, not just directional bias.

All prediction targets are treated as classification problems. For each target, a logistic regression classifier is trained on signals derived from model outputs (sentiment scores, extracted guidance values, uncertainty markers, and QA-derived features). Holding the prediction architecture constant across all language model conditions ensures that prediction differences are attributable to language understanding quality.

2.3 Datasets

The **lamini/earnings-calls-qa** corpus (approximately 860,000 transcript QA pairs) provides training data for Tasks 1 and 2. The **Bose345/sp500_earnings_transcripts** corpus provides transcripts for prediction experiments, covering tens of thousands of calls across S&P 500 firms from 2010 to 2024. **Financial PhraseBank** (Malo et al., 2014) provides 4,840 annotated financial sentences for Task 3. **FinanceBench** (Islam et al., 2023) provides QA pairs over public company 10-K, 10-Q, and 8-K filings and serves as our cross-document-domain evaluation benchmark. **FinQA** (Chen et al., 2021) and **ConvFinQA** (Chen et al., 2022) provide additional numerical reasoning evaluation. Stock direction and volatility labels are computed from historical prices using *yfinance*. Earnings surprise labels are computed using analyst EPS consensus estimates from *yahooquery*. All data splits are chronological to prevent leakage.

3 Component A: Fine-Tuning Strategies and Architecture Comparison

3.1 Model Families

We evaluate three architecture families to determine whether fine-tuning benefits are architecture-

specific or general across inductive biases.

Encoder-only: *ProsusAI/finbert* (Araci, 2019), a BERT-based model pretrained on financial corpora. We fine-tune with task-specific classification and span extraction heads. This model represents strong financial domain pretraining with a discriminative architecture.

Encoder-decoder: *google/flan-t5-base* (Chung et al., 2022), fine-tuned with task-prefixed input formatting. This allows direct evaluation of a generative seq2seq architecture on extraction and QA without the distributional mismatch of prompting-based generation.

Decoder-only: A publicly available instruction-tuned LLM (7B parameters) fine-tuned with LoRA (Hu et al., 2022) at rank $r \in \{4, 16, 64\}$. We also evaluate the base model with prompting only to isolate fine-tuning effects.

3.2 Fine-Tuning Regimes

For each model family, we compare three training regimes:

Single-task fine-tuning. Each model is trained independently on each of the three understanding tasks. This provides baseline per-task performance and establishes the upper bound for task-specific optimization.

Joint multi-task fine-tuning. All three tasks are trained simultaneously with uniform or temperature-scaled task sampling. This tests whether financial language understanding tasks benefit from shared representations.

Curriculum fine-tuning. Tasks are presented in a fixed order designed to build from simpler (sentiment) to more complex (QA) linguistic demands. We test two orderings: sentiment $\rightarrow$ extraction $\rightarrow$ QA and extraction $\rightarrow$ sentiment $\rightarrow$ QA.

3.3 Ablation Studies

Within the decoder-only family, we ablate LoRA rank ($r \in \{4, 16, 64\}$), training data size (10%, 25%, 50%, 100% of available examples), and task ordering in curriculum fine-tuning. Within the encoder-only family, we ablate layer freezing (freeze bottom $k \in \{0, 6, 9\}$ transformer blocks). These ablations form a secondary contribution documenting the sensitivity of financial fine-tuning to engineering choices.

3.4 Cross-Document-Domain Transfer

All fine-tuned models are evaluated zero-shot on FinanceBench, which contains QA over SEC fil-

ings, without any SEC-specific training. This tests whether financial understanding trained on earnings transcripts generalizes to regulatory documents. We additionally fine-tune on a 50/50 mixed corpus of transcripts and SEC filings and compare zero-shot cross-domain performance before and after mixed training.

4 Component B: Retrieval-Augmented Generation for Financial QA

4.1 Motivation

Fine-tuning encodes domain knowledge in model weights, which is static after training. Retrieval-augmented generation (RAG) encodes knowledge in an external index that can be updated without retraining. For financial applications, where facts change quarterly and documents are long, RAG may be a more practical architecture. Component B investigates whether retrieval can substitute for or complement parametric fine-tuning on the QA task.

4.2 Retrieval Strategies

We implement and compare three retrieval strategies:

Sparse retrieval (BM25). A classical bag-of-words retrieval using BM25 scoring over transcript passage indices. This serves as a strong lexical baseline and is particularly relevant for retrieval of numerical entities, which have low paraphrase variation.

Dense retrieval. Passages are embedded using sentence-transformers/all-MiniLM-L6-v2 (Reimers and Gurevych, 2019) and indexed using FAISS. Query embeddings are computed at inference time, and top- k passages are retrieved by cosine similarity.

Hybrid retrieval. BM25 and dense scores are combined via linear interpolation with a tunable mixture weight α . We search over $\alpha \in \{0.2, 0.4, 0.6, 0.8\}$ on a development set.

For all strategies, we evaluate top- k retrieval recall at $k \in \{1, 3, 5, 10\}$ against gold supporting spans. Retrieved passages are prepended as context for the reader model.

4.3 Reader Models and RAG Conditions

We evaluate four experimental conditions on the QA task:

1. Base LLM with no retrieval and no fine-tuning (zero-shot prompting only)

2. Base LLM with the best retrieval strategy (RAG only)

3. Fine-tuned LLM with no retrieval (fine-tuning only)

4. Fine-tuned LLM with the best retrieval strategy (RAG + fine-tuning)

This 2×2 factorial design isolates retrieval and fine-tuning effects and tests whether they are additive, redundant, or complementary.

4.4 Retrieval Failure Analysis

For cases where the system produces incorrect answers, we distinguish two failure modes: (1) retrieval failure, where the correct supporting passage was not retrieved; and (2) reader failure, where the correct passage was retrieved but the answer was still wrong. This decomposition identifies whether engineering effort should be directed toward better retrieval or better reading comprehension. We report failure mode breakdown by question type (numerical, factual, guidance-related) and document section.

4.5 Long-Document Retrieval

Earnings call transcripts can exceed 10,000 tokens. We study the effect of passage chunking granularity (sentence, 128-token, 256-token, 512-token windows) on retrieval recall and end-to-end QA accuracy, providing practical guidance for applying RAG to long financial documents.

5 Component C: Predictive Transfer and Robustness Evaluation

5.1 Feature Extraction and Signal Construction

For each model condition evaluated in Components A and B, we extract a fixed set of signals from transcripts: (1) overall sentiment score and section-level sentiment scores (prepared remarks, analyst Q&A separately); (2) numerical guidance signals (extracted revenue/EPS guidance values, year-over-year change direction); (3) uncertainty markers (hedging language frequency, modal verb counts, extracted from QA and numerical extraction outputs); and (4) QA-derived confidence signals (answer length, citation coverage, hallucination rate as a proxy for model uncertainty). These signals are constructed identically across all model

conditions, ensuring that prediction differences reflect language understanding rather than feature engineering.

5.2 Prediction Models and Evaluation

For each of the three prediction targets (stock direction, earnings surprise, volatility), we train logistic regression classifiers on the extracted signals. We evaluate using AUC-ROC, accuracy, balanced accuracy, and expected calibration error. Confidence intervals are computed via bootstrap resampling with $n = 1,000$ iterations. We test whether fine-tuned models produce significantly higher AUC than baselines using a paired permutation test.

To contextualize our findings, we compare against prior reported AUC values for earnings call prediction (Qin and Yang, 2019). We also compare predictions derived from Q&A sections alone against prepared remarks alone, testing whether spontaneous analyst dialogue contains more predictive signal than scripted management statements.

5.3 Temporal Robustness

Using the chronological corpus, we partition test data into three market-period buckets: pre-COVID (2015–2019), COVID period (2020–2021), and post-COVID (2022–2024). Models are trained only on data prior to each evaluation period. We evaluate all model conditions on each period bucket and plot performance-over-time curves to identify degradation patterns. We additionally conduct vocabulary shift analysis to identify financial terminology that emerges in later periods (e.g., pandemic-era hedging language, supply chain terminology) and examine whether models trained on earlier periods handle novel vocabulary.

5.4 Sector-Stratified and Firm-Size Stratified Analysis

We stratify all prediction results by GICS sector (technology, healthcare, financials, industrials, and others) and by market capitalization tier (large cap vs. mid cap). This analysis tests whether language understanding improvements are concentrated in particular firm types—for example, whether fine-tuning benefits are larger for firms in sectors with more idiosyncratic language (healthcare regulatory language, technology forward guidance) than sectors with more standardized reporting.

5.5 Structured Error Analysis

We implement a structured error categorization pipeline across all tasks and prediction targets. For extraction, errors are labeled as: wrong number, wrong entity, hallucinated value, or missed mention. For QA, errors are labeled as: wrong span, unsupported citation, hallucination, or correct answer with wrong citation. For prediction, errors are analyzed along four dimensions: prediction target, sector, market cap tier, and model condition. Error profiles are compared across model families and fine-tuning regimes to identify whether architectural or training choices produce systematically different failure modes.

6 Baselines

We evaluate against four baselines of increasing linguistic sophistication: (1) majority class prediction; (2) Loughran-McDonald financial sentiment dictionary (Loughran and McDonald, 2011) applied to full transcript text; (3) zero-shot FinBERT; and (4) base instruction-tuned LLM without fine-tuning or retrieval. These baselines are evaluated identically to all fine-tuned conditions across all three prediction targets.

7 Related Work

Lexicon-based financial sentiment analysis (Loughran and McDonald, 2011) and domain-pretrained models such as FinBERT (Araci, 2019) are the dominant approaches to financial NLP. Prior work shows earnings call language predicts market behavior (Qin and Yang, 2019), but transcripts are typically treated as flat documents. Numerical reasoning datasets such as FinQA (Chen et al., 2021) and ConvFinQA (Chen et al., 2022) evaluate financial comprehension but do not connect understanding to prediction. Large language model studies such as BloombergGPT (Wu et al., 2023) examine financial pretraining at scale but do not systematically compare architecture families or fine-tuning strategies. Retrieval-augmented generation has been studied in open-domain settings (Lewis et al., 2020), but its relative utility compared to fine-tuning in financial document domains, and the effect of retrieval strategy choice, have not been characterized. Multi-task learning for NLP understanding tasks has been studied broadly (Raffel et al., 2020) but not specifically for the relationship between financial task co-training and downstream predictive transfer.

8 Work Plan

Component A: Architecture and Fine-Tuning.

Implements the full data pipeline: transcript pre-processing, section segmentation, SEC filing extraction from FinanceBench, training example construction, and chronological splits. Fine-tunes all three model families across all three tasks and three training regimes. Runs LoRA rank and layer-freezing ablations. Conducts cross-document-domain transfer experiments. Leads Component A evaluation and result tables.

Component B: Retrieval-Augmented Generation.

Implements the RAG pipeline: passage chunking, BM25 indexing (via rank_bm25), dense embedding indexing (via FAISS and sentence-transformers), and hybrid retrieval with interpolation. Evaluates all retrieval conditions on the QA task. Implements the retrieval failure analysis decomposition. Conducts the chunking granularity ablation. Leads Component B evaluation and result tables.

Component C: Prediction and Robustness. Implements the signal extraction pipeline and logistic regression prediction classifiers for all three prediction targets. Computes bootstrap confidence intervals and permutation test statistics. Conducts temporal robustness evaluation, sector-stratified analysis, and firm-size stratified analysis. Designs and implements the structured error analysis pipeline. Leads Component C evaluation, result tables, and error analysis.

Integration and shared milestones. All team members contribute to a unified evaluation framework that connects outputs from Components A and B to Component C, with integration work shared across the team. Given the timeline, the project will proceed in overlapping phases over the course of a month: the first phase focuses on setting up the data pipeline, shared infrastructure, and baseline implementations; the middle phase focuses on running the primary experiments for Components A, B, and C in parallel; and the final phase focuses on targeted ablations, cross-component analysis, error analysis, and writing.

9 References

Dogu Araci. 2019. FinBERT: Financial Sentiment Analysis with Pre-trained Language Models. *arXiv preprint arXiv:1908.10063*.

Zhuang Chen et al. 2021. FinQA: A Dataset of Numerical Reasoning over Financial Data. In *EMNLP*, pages 3697–3711.

Zhuang Chen et al. 2022. ConvFinQA: Exploring the Chain of Numerical Reasoning in Conversational Finance Question Answering. In *EMNLP*, pages 6279–6292.

Hyung Won Chung et al. 2022. Scaling Instruction-Finetuned Language Models. *arXiv preprint arXiv:2210.11416*.

Edward Hu et al. 2022. LoRA: Low-Rank Adaptation of Large Language Models. In *ICLR*.

Zain Ul Islam Fazal et al. 2023. FinanceBench: A New Benchmark for Financial Question Answering. *arXiv preprint arXiv:2311.11944*.

Patrick Lewis et al. 2020. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. In *NeurIPS*.

Tim Loughran and Bill McDonald. 2011. When Is a Liability Not a Liability? *Journal of Finance*, 66(1):35–65.

Pekka Malo et al. 2014. Good Debt or Bad Debt: Detecting Semantic Orientations in Economic Texts. *Journal of the American Society for Information Science and Technology*, 65(4):782–796.

Yao Qin and Yi Yang. 2019. What You Say and How You Say It Matters: Predicting Stock Volatility Using Verbal and Vocal Cues. In *ACL*, pages 390–401.

Colin Raffel et al. 2020. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *Journal of Machine Learning Research*, 21(140):1–67.

Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks. In *EMNLP*, pages 3982–3992.

Shijie Wu et al. 2023. BloombergGPT: A Large Language Model for Finance. *arXiv preprint arXiv:2303.17564*.